

Stein and Shakarchi Complex Analysis Solution Manual

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Chapter 1

9. Show that in polar coordinates CR equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}$$

Use these equations to show that the logarithm function defined by

$$\log z = \log r + i\theta \quad \text{where } z = re^{i\theta} \text{ with } -\pi < \theta < \pi$$

is holomorphic in the region $r > 0$ and $-\pi < \theta < \pi$

Proof. Let $f = u + iv$. Cauchy-Riemann equations tell us $\partial_x u = \partial_y v$, $\partial_y u = -\partial_x v$. In polar coordinates $z = x + iy = r(\cos \theta + i \sin \theta)$ Therefore,

$$\begin{aligned} \partial_r u &= \partial_x u \cdot \partial_r x + \partial_y u \cdot \partial_r y \\ &= \partial_x u \cdot \cos \theta + \partial_y u \cdot \sin \theta \\ &= \partial_y v \cdot \cos \theta - \partial_x v \cdot \sin \theta \\ &= \frac{1}{r} \cdot \partial_\theta v \end{aligned}$$

Similarly, we also obtain

$$\partial_r v = -\frac{1}{r} \partial_\theta u$$

For $\log z$, $u = \log r$ and $v = \theta$. Then, we check CR equations,

$$\frac{\partial u}{\partial r} = 1/r = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \frac{1}{r} \frac{\partial u}{\partial \theta} = 0 = -\frac{\partial v}{\partial r}$$

So $\log z$ is holomorphic. □

17. Show that if $\{a_n\}$ is a sequence of non-zero complex numbers such that $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = L$, then $\lim_{n \rightarrow \infty} |a_n|^{1/n} = L$.

Proof. Let $\varepsilon > 0$ be given. Then for $n > N$, we have

$$\left| L - \frac{|a_{n+1}|}{|a_n|} \right| < \varepsilon \implies (L - \varepsilon)|a_n| < |a_{n+1}| < (L + \varepsilon)|a_n|$$

Multiplying these inequalities from $n = N$ to $n = k > N$, we get

$$(L - \varepsilon)^{k-N} < \frac{|a_k|}{|a_N|} < (L + \varepsilon)^{k-N}$$

For large enough N ,

$$L - 2\varepsilon < (L - \varepsilon)^{1-N/k} |a_N|^{1/k} < |a_k|^{1/k} < |a_N|^{1/k} (L + \varepsilon)^{1-N/k} < L + 2\varepsilon$$

□

22. Show that $\mathbb{N}$ cannot be partitioned into a finite number of subsets that are in arithmetic progression with distinct steps.

Proof. Assume that $\mathbb{N} = A_1 \cup \dots \cup A_k$ where $A_j = \{a_j + nd_j : n \in \mathbb{N}\}$ for some $d_j \in \mathbb{N}, a_j \in \mathbb{N}$. WLOG, we can assume $d_1 < d_2 < \dots < d_k$. WLOG, assume $1 \in A_1$. Then, numbers from 2 to d_1 must be inside different subsets. By renaming, we can assume $i \in A_i$ for $i = 2, \dots, d_1$. Then, $1 + d_1 \in A_1$. Since $d_1 < d_j$, we know that A_j cannot contain a number less than or equal to $j + d_1$. Then, we see that $2 + d_1 \notin A_2$. Say $2 + d_1 \in A_j$ for some $j \in \{3, \dots, k\}$. Then, $2 + d_1 - j < d_1 < d_j$ which is a contradiction. Hence, there is no suitable subset for $2 + d_1$. This shows that $\mathbb{N}$ cannot be covered by finitely many subsets in arithmetic order. □

Chapter 2

4. Prove that for all $\xi \in \mathbb{C}$ we have

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{2\pi i x \xi} dx$$

Proof.

$$-\pi x^2 + 2\pi i x \xi = -\pi(x - i\xi)^2 - \pi\xi^2$$

Moreover,

$$\begin{aligned} \int_{-\infty}^{\infty} e^{-\pi x^2} e^{2\pi i x \xi} dx &= e^{-\pi\xi^2} \int_{-\infty}^{\infty} e^{-\pi(x-i\xi)^2} dx \\ &= e^{-\pi\xi^2} \end{aligned}$$

We know that $\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1$. Let $\xi = a + ib$ then

$$\int_{-\infty}^{\infty} e^{-\pi(x-i\xi)^2} dx = \int_{-\infty}^{\infty} e^{-\pi(x+ia)^2} dx$$

Using the rectangular contour $(-R, R) \rightarrow (R, R + ia) \rightarrow (R + ia, -R + ia) \rightarrow (-R + ia, -R)$ we obtain

$$\int_{-\infty}^{\infty} e^{-\pi x^2} dx = \int_{-\infty}^{\infty} e^{-\pi(x+ia)^2} dx = 1$$

because the integral over the vertical segments cancel each other as $R \rightarrow \infty$. $\square$

5. Suppose f is complex differentiable inside Ω and $T \subset \Omega$ is a triangle whose interior is also contained inside Ω . Apply Green's Theorem to show that

$$\int_T f(z) dz = 0$$

Proof. If $f = u + iv$ and $z = x + iy$ then $dz = dx + idy$. Green's theorem says

$$\int_T F dx + G dy = \int_{\text{int } T} (\partial_x G - \partial_y F) dx dy$$

Then,

$$\begin{aligned} \int_T f(z) dz &= \int_T (u + iv)(dx + idy) \\ &= \int_T u dx - v dy + i \int_T v dx + u dy \\ &= 0 \end{aligned}$$

by Cauchy Riemann equations. $\square$

6. Let Ω be an open subset of $\mathbb{C}$ and let $T \subset \Omega$ be a triangle whose interior is also contained in Ω . Suppose that f is a function holomorphic inside Ω except possibly at a point w inside T . Prove that if f is bounded near w , then

$$\int_T f(z) dz = 0$$

Proof. Create a contour circle around w and then connect the three edges of the triangle to this disc. Then f is holomorphic inside the three curves. $\int_T f = \lim_{\epsilon \rightarrow 0} \int_0^{2\pi} f(w + \epsilon e^{i\theta}) i\epsilon e^{i\theta} d\theta = 0$ $\square$

7. Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic then $2|f'(0)| \leq d$ where $d = \sup_{z, w \in \mathbb{D}} |f(z) - f(w)|$. Moreover, show that the equality holds precisely when f is linear.

Proof. Use

$$f'(0) = \frac{1}{2\pi i} \int_{|z|=r} \frac{f(z)}{z^2} dz = \frac{1}{2\pi i} \int_{|z|=r} \frac{-f(-z)}{z^2} dz$$

So

$$\begin{aligned} 2|f'(0)| &\leq \frac{1}{2\pi} \int_{|z|=r} \left| \frac{f(z) - f(-z)}{z^2} \right| dz \\ &= \frac{d}{r} \rightarrow d \text{ as } r \rightarrow 1. \end{aligned}$$

□

8. If f is a holomorphic function on the strip $-1 < y < 1, x \in \mathbb{R}$ with

$$|f(z)| \leq A(1 + |z|)^\eta, \quad \eta \text{ a fixed real number}$$

for all z in that strip, show that for each integer n , there exists $A_n \geq 0$ so that

$$|f^{(n)}(x)| \leq A_n(1 + |x|)^\eta, \quad \forall x \in \mathbb{R}$$

Proof. Consider Cauchy inequalities, we need to bound $\frac{n!}{(1/2)^n} A \sup_{z \in C_{1/2}(x)} (1 + |z|)^\eta$. If $\eta > 0$, the supremum is attained when $|z| = |x| + 1/2$. If $\eta < 0$, the supremum is attained when $|z| = |x| - 1/2$ for $|x| > 1/2$ and $|z| = 1/2 - |x|$ for $|x| \leq 1/2$. □

9. To be written. (Showed up in midterm).

10. Probably uses Runge's approximation theorem.

11. Computation heavy, skipped.

12. Let u be a real-valued harmonic C^2 function defined on the unit disc. Prove that there exists a holomorphic function f on the unit disc such that $\Re(f) = u$. Also show that the imaginary part of f is uniquely defined up to an additive constant.

Proof. Since $\partial_z f = 2\partial_z u$ put $g(z) = 2\partial_z u$. We have $g(z) = 2u_x - 2iu_y$. Then, $(u_x)_x = -(u_y)_y$ and $(u_x)_y = -(-u_y)_x$ so g satisfies CR equalities, hence g is holomorphic. Then, there exists $F : \mathbb{D} \rightarrow \mathbb{C}$ with $F' = g$. Say $\Re(F) = U$ then $F'(z) = 2\partial_z U = g(z) = 2\partial_z u \implies \partial_z U = \partial_z u \implies U = u + c$ for some constant $c \in \mathbb{R}$. Then $u = \Re(F - c)$. The imaginary part is given by

$$v(z) = \int_\gamma u_x dy - \int_\gamma u_y dx + C$$

where $C = v(0) \in \mathbb{R}$ and γ is any curve connecting 0 to z . □

13. Suppose f is analytic in $\mathbb{C}$ such that for each z_0 at least one coefficient in the expansion $f(z) = \sum c_n(z - z_0)^n$ is zero. Prove that f is a polynomial.

Proof. Define a map $g : \mathbb{C} \rightarrow \mathbb{N}$ given by $g(z_0) = m$ where m is the smallest number satisfying $f^{(m)}(z_0) = 0$. Since $c_n = f^{(n)}(z_0)/n!$, $g(z_0) = m \implies c_m = 0$. It follows that $c_n = 0$ for all $n \geq m$. We claim that there exists $m \in \mathbb{N}$ whose preimage $g^{-1}(m)$ is an uncountable set. If this is not the case, then the cardinality of $\mathbb{C}$ would be $\sum_{n=0}^{\infty} |g^{-1}(n)|$ which is countably infinite. However, $\mathbb{C}$ is uncountable. Thus, there must exist $m \in \mathbb{N}$ with $g^{-1}(m)$ an uncountable set. Then, $f^{(m)}(z) = 0$ for uncountably many z . Hence, the zeros of $f^{(n)}$ cannot be isolated because a discrete set in $\mathbb{C}$ must be countable. This shows that $f^{(m)}(z) = 0$ for all $z \in \mathbb{C}$. Thus, f is a polynomial of degree at most $m - 1$. $\square$

14. Suppose $f(z)$ is holomorphic in an open set containing the closed unit disc except for a pole at z_0 on the unit circle. Show that if $f(z) = \sum a_n z^n$ denotes the power series expansion of f on the unit disc, then $\lim_{z \rightarrow z_0} \frac{a_n}{a_{n+1}} = z_0$.

Proof. Write

$$f(z) = \frac{b_{-k}}{(z - z_0)^k} + \cdots + \frac{b_{-1}}{(z - z_0)} + G(z)$$

near z_0 . Then,

$$\sum a_n z^n = \frac{b_{-k}}{(z - z_0)^k} + \cdots + \frac{b_{-1}}{(z - z_0)} + \sum_n c_n z^n$$

Since

$$\frac{1}{(z - z_0)^l} = \sum_{n=0}^{\infty} (-1)^l \binom{l+n-1}{l-1} z_0^{-n-l} z^n$$

for $|z| < 1$, we get

$$\sum a_n z^n = \sum_{l=1}^k \sum_{n=0}^{\infty} (-1)^l \binom{l+n-1}{l-1} z_0^{-l-n} z^n + \sum_{n=0}^{\infty} c_n z^n$$

Note that

$$\lim_{n \rightarrow \infty} (-1)^l \binom{l+n-1}{l-1} = \frac{1}{(l-1)!} n^{l-1}$$

This means

$$a_n = c_n + \sum_{l=1}^k (-1)^l \binom{l+n-1}{l-1} z_0^{-l-n}$$

Therefore

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} &= \lim_{n \rightarrow \infty} \frac{c_n + \sum_{l=1}^k (-1)^l \binom{l+n-1}{l-1} z_0^{-l-n}}{c_{n+1} + \sum_{l=1}^k (-1)^l \binom{l+n}{l-1} z_0^{-l-n-1}} \\ &= \lim_{n \rightarrow \infty} \frac{c_n z_0^n + \sum_{l=1}^k \frac{1}{(l-1)!} n^{l-1} z_0^{-l}}{c_{n+1} z_0^n + \sum_{l=1}^k \frac{1}{(l-1)!} (n+1)^{l-1} z_0^{-l-1}} \\ &= z_0 \end{aligned}$$

For the last equality, divide both nominator and denominator by n^{k-1} and note that the first terms of both go to zero while only the last term in the sum does not vanish for both, leaving z_0^{-l}/z_0^{-l-1} as the answer. $\square$

15. Suppose f is a non-vanishing continuous function on $\overline{\mathbb{D}}$ that is holomorphic in $\mathbb{D}$. Prove that if

$$|f(z)| = 1 \quad \text{whenever} \quad |z| = 1$$

then f is constant.

Proof. Define

$$F(z) = \begin{cases} f(z) & |z| \leq 1 \\ \frac{1}{f(\frac{1}{\bar{z}})} & |z| > 1 \end{cases}$$

Then, since f is non-vanishing in $\overline{\mathbb{D}}$, F is well-defined. Also, f is bounded both from below and above. Say $m < f(z) < M$ for all $z \in \overline{\mathbb{D}}$. Then, $1/M < |F(z)| < 1/m$ for all $|z| > 1$ and $|F(z)| < M$ for $|z| \leq 1$. So F is bounded. We show that F is holomorphic by using Morera's theorem. Let T be a triangle whose interior is outside $\mathbb{D}$. Then,

$$\int_T F(z) = \int_{T' \subset \mathbb{D}} 1/\overline{f(1/\bar{z})} dz = 0$$

there $T' = \{z \in \mathbb{D} : 1/\bar{z} \in T\}$ is a toy contour inside $\mathbb{D}$. Since $1/\overline{f(1/\bar{z})}$ is holomorphic on $\mathbb{D}$, this integral is 0 by Cauchy's theorem. If T is a triangle that intersects $\mathbb{D}$ and its exterior then construct two toy contours and move them a tiny bit away from the boundary $\partial\mathbb{D}$ so that their integral is 0. By continuity, the integral of F over this triangle is 0 as well. Hence F is holomorphic. By Liouville's theorem F is constant, hence f is constant as well. $\square$

Chapter 3

1. skipped.

2. Evaluate

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^4}$$

Proof. We use the semi-circle of radius R as the contour. $(z^4+1) = (z^2-i)(z^2+i) = (z - e^{i\pi/4})(z + e^{i\pi/4})(z - e^{-i\pi/4})(z + e^{-i\pi/4})$ each zero is of order 1 so we compute the residues of poles of $1/(z^4+1)$ inside the semicircle of radius $R > 1$.

$$\begin{aligned} \lim_{z \rightarrow e^{i\pi/4}} (z - e^{i\pi/4}) \frac{1}{z^4 + 1} &= \frac{e^{-i\pi/4}}{4i} \\ \lim_{z \rightarrow -e^{-i\pi/4}} (z + e^{-i\pi/4}) \frac{1}{z^4 + 1} &= \frac{e^{i\pi/4}}{4i} \end{aligned}$$

So

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{1+x^4} dx = 2\pi i \frac{e^{i\pi/4} + e^{-i\pi/4}}{4i} = \frac{\sqrt{2}\pi}{2}$$

The integral over the arc goes to zero since

$$\left| \int_0^{2\pi} \frac{Rie^{i\theta}}{1+R^4e^{4i\theta}} d\theta \right| \leq \int_0^{2\pi} \frac{R}{|1+R^4e^{4i\theta}|} d\theta \rightarrow 0$$

as $R \rightarrow \infty$. □

3. Show that

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2} dx = \pi \frac{e^{-a}}{a} \quad \text{for } a > 0.$$

Proof. Since we need to get $\frac{\cos x}{x^2 + a^2}$ we want to integrate $\frac{e^{iz}}{z^2 + a^2}$ and take the real part.

Use the semicircle contour of radius $R > a$. The poles of $\frac{e^{iz}}{z^2 + a^2}$ are at $z = ia$ and $z = -ia$. The second pole is not inside the upper semicircle of radius R . So

$$\lim_{z \rightarrow ia} (z - ia) \frac{e^{iz}}{z^2 + a^2} = \frac{e^{-a}}{2ia}$$

By the residue theorem

$$\int_{-R}^R \frac{e^{ix}}{x^2 + a^2} dx + \int_0^\pi \frac{e^{iRe^{i\theta}}}{R^2e^{2i\theta} + a^2} d\theta = 2\pi i \cdot \frac{e^{-a}}{2ia} = \pi \frac{e^{-a}}{a}.$$

Moreover,

$$\left| \int_0^\pi \frac{e^{iRe^{i\theta}}}{R^2e^{2i\theta} + a^2} d\theta \right| \leq \int_0^\pi \frac{e^{-R \sin \theta}}{|R^2e^{2i\theta} + a^2|} d\theta \xrightarrow{R \rightarrow \infty} 0$$

Hence,

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2 + a^2} dx = \pi \frac{e^{-a}}{a}$$

Taking the real parts, we see that

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2} dx = \pi \frac{e^{-a}}{a}$$

Taking the imaginary part also shows that

$$\int_{-\infty}^{\infty} \frac{\sin x}{x^2 + a^2} dx = 0$$

as expected since it is an odd function. □

4. Show that

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + a^2} dx = \pi e^{-a}$$

Proof. Apply the residue theorem to the integral of the function $\frac{ze^{iz}}{z^2+a^2}$ over the upper semicircle of radius R . Let $R \rightarrow \infty$, then the integral over the arc goes to zero and the residue theorem gives $i\pi e^{-a}$. The desired integral is the imaginary part of the integral of $\frac{ze^{iz}}{z^2+a^2}$. $\square$

5. Show that

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i \zeta x}}{(1+x^2)^2} dx = \frac{\pi}{2} (1 + 2\pi |\zeta|) e^{-2\pi |\zeta|}$$

for every $\zeta \in \mathbb{R}$.

Proof. For $\zeta > 0$, use the lower semicircle of radius R . The pole is at $-i$ with residue

$$\lim_{z \rightarrow -i} \left[(z+i)^2 \frac{e^{-2\pi i \zeta z}}{(z^2+1)^2} \right]' = \frac{i}{4} e^{-2\pi \zeta} (2\pi \zeta + 1)$$

Then, the integral over the lower arc goes to zero since

$$\begin{aligned} \left| \int_0^{-\pi} \frac{e^{-2\pi i \zeta R e^{i\theta}}}{(1+R^2 e^{2i\theta})^2} R i e^{i\theta} d\theta \right| &\leq \int_0^{-\pi} \frac{e^{2\pi \zeta R \sin \theta} R}{R^2} d\theta \\ &\leq \int_0^{-\pi} \frac{1}{R} d\theta = \pi/R \rightarrow 0 \end{aligned}$$

as $R \rightarrow \infty$. Notice the crucial reason we choose the lower semicircle is that $2\pi \zeta R \sin \theta < 0$ for $\theta \in [-\pi, 0]$ and $\zeta > 0$ so $e^{2\pi \zeta R \sin \theta} < 1$ $\square$

9. Show that

$$\int_0^1 \log(\sin(\pi x)) dx = -\log 2$$

Proof. Notice

$$\begin{aligned} \log(1 - e^{2\pi i z}) &= \log(e^{\pi i z} (e^{-\pi i z} - e^{\pi i z})) \\ &= \pi i z + \log((-2i) \sin(\pi z)) \\ &= \pi i z + \log 2 - i\pi/2 + \log(\sin \pi z) \end{aligned}$$

We integrate this over the contour $(\varepsilon, 1-\varepsilon) \rightarrow 1 + \varepsilon e^{i\theta} \rightarrow (1 + \varepsilon i, 1 + Ri) \rightarrow (1 + Ri, Ri) \rightarrow (Ri, \varepsilon i) \rightarrow \varepsilon e^{i\theta}$. The integral over the vertical lines cancel each other. The integral at the top is,

$$\int_0^1 \left| \log(1 - e^{2\pi i (Ri+t)}) \right| dt \sim \int_0^1 e^{-2\pi R} \cdot |e^{2\pi i t}| + O(e^{2R}) dt \xrightarrow{R \rightarrow \infty} 0$$

$\square$

10. Show that

$$\int_0^\infty \frac{\log x}{x^2 + a^2} dx = \frac{\pi}{2a} \log a$$

Proof. Let $f(z) = \frac{\log z}{z^2 + a^2}$. Use the indented upper semicircle centered at the origin. Then, the pole is at ia and the residue is

$$\lim_{z \rightarrow ia} (z - ia) \frac{\log z}{z^2 + a^2} = \frac{\pi \log a}{a} + i \frac{\pi^2}{2a}$$

Γ_1 is the small semicircle of radius ε .

$$\begin{aligned} \left| \int_0^\pi \frac{\log(\varepsilon e^{i\theta})}{\varepsilon^2 e^{2i\theta} + a^2} \varepsilon i e^{i\theta} d\theta \right| &\leq \int_0^\pi \frac{|\log(\varepsilon e^{i\theta})| \varepsilon}{|\varepsilon^2 e^{2i\theta} + a^2|} \theta \\ &\xrightarrow{\varepsilon \rightarrow 0} \int_0^\pi \frac{|\log \varepsilon + i\theta| \varepsilon}{a^2} d\theta \\ &\xrightarrow{\varepsilon \rightarrow 0} 0 \end{aligned}$$

Γ_3 is the big semicircle of radius R .

$$\begin{aligned} \left| \int_0^\pi \frac{\log(Re^{i\theta})}{R^2 e^{2i\theta} + a^2} R i e^{i\theta} d\theta \right| &\leq \int_0^\pi \frac{R |\log R + i\theta|}{|R^2 e^{2i\theta} + a^2|} d\theta \\ &\leq \int_0^\pi R \frac{\log R + \pi}{R^2 - a^2} d\theta \\ &\xrightarrow{R \rightarrow \infty} 0 \end{aligned}$$

Γ_4 is the part of the curve that is on the negative x -axis.

$$\begin{aligned} \int_{-R}^0 \frac{\log(r)}{r^2 + a^2} dr &= \int_{-R}^0 \frac{\log(-x) + i\pi}{x^2 + a^2} dx \\ &= \int_0^R \frac{\log x + i\pi}{x^2 + a^2} dx \\ &= \int_0^R \frac{\log x}{x^2 + a^2} + i \frac{\pi}{a} \arctan(x/a) \Big|_0^R \\ &\xrightarrow{R \rightarrow \infty} \int_0^\infty \frac{\log x}{x^2 + a^2} dx + i \frac{\pi^2}{2a} \end{aligned}$$

This contour, together with the residue formula gives

$$2I = \frac{\pi \log a}{a} \implies I = \frac{\pi \log a}{2a}$$

□

11. Show that if $|a| < 1$, then

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = 0$$

Proof. Notice that

$$\log |1 - ae^{i\theta}| = \Re(\log(1 - ae^{i\theta}))$$

Therefore,

$$\int_0^{2\pi} \log |1 - ae^{i\theta}| d\theta = \Re \left(\int_0^{2\pi} \log(1 - ae^{i\theta}) d\theta \right)$$

Let $f(a) = \int_0^{2\pi} \log(1 - ae^{i\theta}) d\theta$. Since $\log(1 - ae^{i\theta})$ is a holomorphic function of a for fixed θ and is continuous on $\mathbb{D} \times [0, 2\pi]$ f is a holomorphic function. Now notice

$$f'(a) = \int_0^{2\pi} \frac{-e^{i\theta}}{1 - ae^{i\theta}} d\theta = \int_C \frac{i}{1 - az} dz = 0$$

by Cauchy's theorem since the integrand is holomorphic inside the unit disc, since $|a| < 1$. Then $f(a)$ is a constant function. Therefore

$$\int_0^{2\pi} \log(1 - ae^{i\theta}) d\theta = f(a) = f(0) = 0$$

□

12. Suppose u is not an integer. Prove that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(u+n)^2} = \frac{\pi^2}{\sin(\pi u)^2}$$

Proof. We will integrate $f(z) = \frac{\pi \cot(\pi z)}{(u+z)^2}$ over a circle of radius $N + 1/2$ for a fixed $N \in \mathbb{N}$ and then we let $N \rightarrow \infty$. One pole is at the integers and the other pole is at $-u$.

$$\begin{aligned} \text{Res}(f, n) &= \lim_{z \rightarrow n} (z - n) \frac{\pi \cot(\pi z)}{(u+z)^2} = \frac{1}{(u+n)^2} \\ \text{Res}(f, -u) &= \lim_{z \rightarrow -u} \left[(z + u)^2 \frac{\pi \cot(\pi z)}{(u+z)^2} \right] = \frac{-\pi^2}{\sin^2(\pi u)} \end{aligned}$$

Therefore, we obtain

$$\int_{-\infty}^{\infty} f(z) dz = 2\pi i \left(\sum_{n=-\infty}^{\infty} \frac{1}{(u+n)^2} - \frac{\pi^2}{\sin(\pi u)^2} \right)$$

The integral is zero since

$$\left| \int_0^{2\pi} \frac{\pi \cot(Re^{i\theta})}{(u + Re^{i\theta})^2} Rie^{i\theta} d\theta \right| \leq \int_0^{2\pi} \frac{\pi |\cot(Re^{i\theta})| R}{|u + Re^{i\theta}|^2} d\theta$$

Since

$$|\cot(z)| = \frac{|e^{iz} + e^{-iz}|}{|e^{iz} - e^{-iz}|} < \frac{e^{-y} + e^y}{|e^{-y} - e^y|} \xrightarrow{N \rightarrow \infty} 1$$

This means there exists $M \in \mathbb{R}$ such that for $N > M$, $|\cot(\pi z)| < 2$ for $z \in C_N$. Then,

$$\begin{aligned} \left| \int_{-\infty}^{\infty} f(z) dz \right| &\leq 2\pi \int_0^{2\pi} \frac{R}{|u + Re^{i\theta}|^2} d\theta \\ &\leq 2\pi \int_0^{2\pi} \frac{R}{(R - |u|)^2} d\theta \xrightarrow{R \rightarrow \infty} 0 \end{aligned}$$

Therefore we see that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(u+n)^2} = \frac{\pi^2}{\sin(\pi u)^2}$$

□

13. Suppose f is holomorphic on the punctured disc $D_r(z_0) - z_0$ and satisfies

$$|f(z)| \leq A|z - z_0|^{-1+\epsilon}$$

for some $\epsilon > 0$ for all z near z_0 . Show that f has a removable singularity at z_0 .

Proof. Put $g(z) = (z - z_0)f(z)$ then $|g(z)| \leq A|z - z_0|^\epsilon \leq Ar^\epsilon$ so by Riemann's theorem on removable singularities, $g(z)$ has a removable singularity at z_0 . Then, f can't have an essential singularity at z_0 . If f has a pole at z_0 then we can write

$$f(z) = \frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{a_{-1}}{z - z_0} + G(z)$$

in a neighbourhood of z_0 . But now

$$|g(z)| = \left| \frac{a_{-n}}{(z - z_0)^{n-1}} + \cdots + a_{-1} + (z - z_0)G(z) \right| \leq A|z - z_0|^\epsilon \xrightarrow{z \rightarrow z_0} 0$$

which is only possible if $f(z) = G(z)$. Hence, f has a removable singularity at z_0 . □

14. **POTENTIAL** Prove that all entire functions that are injective on the complex plane are of the form $f(z) = az + b$ for some $a, b \in \mathbb{C}$.

Proof. Consider $g(z) = f(1/z)$. If $g(z)$ has a removable singularity at 0, then we can define it to be holomorphic at 0. If $g(z) = \sum_{n=0}^{\infty} a_n z^n$ denotes the holomorphic extension, we would have $f(z) = \sum_{n=0}^{\infty} a_n z^{-n}$ but this is impossible since $f(z) = \sum_{n=1}^{\infty} b_n z^n$ for some constants $b_n \in \mathbb{C}$.

If $g(z)$ has a pole of order n at 0, then

$$g(z) = \frac{a_{-n}}{z^n} + \cdots + \frac{a_{-1}}{z} + \sum_{n=0}^{\infty} a_n z^n$$

and so

$$f(z) = a_{-n}z^n + \cdots + a_{-1}z + a_0$$

we see that we must have $a_n = 0$ for $n > 0$ by the fact that f is entire on $\mathbb{C}$, so it cannot have a principal part. In this case, for f to be injective, it has to be a polynomial of degree 1.

Finally, if $g(z)$ has an essential singularity at 0 then $g(D_r(0) \setminus \{0\})$ is dense in $\mathbb{C}$ for some $r \in \mathbb{R}$. This means, that $f(\mathbb{C} \setminus D_{1/r}(0))$ is dense in $\mathbb{C}$. Let $0 < \epsilon < 1/r$. Since f is an open map, there exists a $\delta > 0$ such that $D_\delta(f(0)) \subset f(D_\epsilon(0))$. By density, there exists a point w with $|w| > 1/r$ and $|f(w) - f(0)| \leq \delta$ which implies $f(w) \in f(D_\epsilon(0))$ then there exists $z \in D_\epsilon(0)$ such that $f(w) = f(z)$. This is a contradiction since $|z| < \epsilon < 1/r < |w|$ so it contradicts with injectivity of f . Therefore, $f(z) = az + b$. $\square$

15. a) **Cauchy Inequality** Prove that if an entire function satisfies $\sup_{|z|=R} |f(z)| \leq AR^k + B$ then f is a polynomial of degree $\leq k$.

Proof.

$$|f^{(n)}(z)| \leq \frac{n! \|f\|_C}{R^n} \leq \frac{n!(AR^k + B)}{R^n} \xrightarrow{R \rightarrow \infty} 0$$

whenever $n > k$. $\square$

b) **Maximum Modulus Principle** If f is holomorphic in the unit disc, bounded and converges uniformly to zero on the sector $\theta < \arg z < \varphi$ as $|z| \rightarrow 1$, then $f = 0$.

Proof. Let $\alpha \in (\theta, \varphi)$ and let m be the smallest integer with $m\alpha > 2\pi$. Then define $g(z) = f(z)f(ze^{i\alpha}) \dots f(ze^{im\alpha})$. Given $\varepsilon > 0$, there exists $r \in \mathbb{R}$ such that $\forall |z| > r$ we have

$$|f(ze^{it})| < \varepsilon$$

for all $t \in (\theta, \varphi)$. Then, it follows that $g(z)$ converges uniformly to zero on the boundary since given ε , there exists $z = re^{it}$ and $k \in \mathbb{N}$ such that $t + k\alpha \in (\theta, \varphi)$ so that $|f(ze^{ik\alpha})| < \varepsilon \implies |g(z)| < M^m \varepsilon$ for all z with $|z| > r'$ where M is the bound on f . Then, by the maximum modulus principle $g(z) = 0$ on $\mathbb{D}$. In particular, $f(0) = 0$. Now we want to construct a sequence of points z_n with $f(z_n) = 0$ and $z_n \rightarrow 0 \in \mathbb{D}$. Then, the zeros of f are not isolated, in particular f must be the zero function. Consider

$$g\left(\frac{1}{n}\right) = f\left(\frac{1}{n}\right)f\left(\frac{1}{n}e^{i\alpha}\right) \dots f\left(\frac{1}{n}e^{im\alpha}\right) = 0$$

Then, for each n , there exists $0 \leq k_n \leq m$ with $f\left(\frac{1}{n}e^{ik_n\alpha}\right) = 0$. Put $z_n = \frac{1}{n}e^{ik_n\alpha}$. Then, $f(z_n) = 0$ and $|z_n| \rightarrow 0 \implies z_n \rightarrow 0$ as $n \rightarrow \infty$. Hence $f \equiv 0$ on $\mathbb{D}$. $\square$

c) **Maximum Modulus Principle and Cauchy Inequalities** Let $w_1, \dots, w_n$ be points on the unit circle. Prove that there exists a point z in the unit circle such that the product of the distances between z and w_i is at least 1. Also show that it attains the value 1 for some z on the unit circle.

Proof. Define $f(z) = (z - w_1) \dots (z - w_n)$. Then, $f^{(n)}(z) = n!$ is a holomorphic function on the unit disc and continuous on the closed unit disc. By max mod principle,

$$n! \leq \sup_{z \in C} |f^{(n)}(z)| \leq \frac{n!}{1} \sup_{z \in C} |f(z)| \implies 1 \leq |f(z)| \quad \forall z \in \mathbb{D}$$

and since $f(z)$ is continuous on $\overline{\mathbb{D}}$ it attains its minimum on the circle. Therefore, $f(z) = 1$ for some z on the unit circle. $\square$

d) If the real part of an entire function f is bounded, then f is constant.

Proof. Let $f(z) = u(z) + iv(z)$. $u(z)$ is holomorphic since $u'(z) = f'(z)/2$. Since f is bounded, u is bounded. By Liouville's theorem, $u(z)$ is constant. By Cauchy Riemann equations, $v_x = -u_y, v_y = u_x$, in particular $v(z)$ is also a constant. Hence f is a constant. $\square$

16. Suppose f and g are holomorphic in a region containing the disc $|z| \leq 1$. Suppose that f has a simple zero at $z = 0$ and vanishes nowhere else in $|z| \leq 1$. Let

$$f_\epsilon(z) = f(z) + \epsilon g(z)$$

Show that if ϵ is sufficiently small, then

(a) $f_\epsilon(z)$ has a unique zero in $|z| \leq 1$.

Proof. If $f(z)$ does not vanish on $|z| = 1$, then $|f(z)| > N$ for some $N \in \mathbb{R}$ on C . Since $g(z)$ is continuous on $|z| \leq 1$, $|g(z)| \leq M$ for some $M \in \mathbb{R}$ for all $|z| \leq 1$. Next, choose $\epsilon > 0$ small enough such that $N > \epsilon M$. Then, $f_\epsilon(z) \neq 0$ on C since $f_\epsilon(z) = 0 \iff f(z) = -\epsilon g(z) \implies |f(z)| = \epsilon M > N$ a contradiction. Then,

$$n_\epsilon = \int_C \frac{f'_\epsilon(z)}{f_\epsilon(z)} dz$$

is the number of zeros of $f_\epsilon(z)$ inside $\mathbb{D}$. Since f_ϵ is a continuous function of ϵ , n_ϵ is a continuous function too and it is also an integer. Hence it is constant. For $\epsilon = 0$ we have $n_0 = 1$. Thus, $n_\epsilon = 1$ as well, meaning $f_\epsilon(z)$ has a unique zero inside $\mathbb{D}$. $\square$

(b) If z_ϵ denotes the unique zero of $f_\epsilon(z)$ inside $\mathbb{D}$, then $\epsilon \mapsto z_\epsilon$ is continuous.

Proof. Fix $\epsilon_0 > 0$ small enough such that $f_{\epsilon_0}(z)$ has a unique root $z_{\epsilon_0} \in \mathbb{D}$. Given $\zeta > 0$, choose $\delta > 0$ such that

$$|f_{\epsilon_0}(z)| > |\epsilon - \epsilon_0| \cdot |g(z)|$$

for every $z \in C_\zeta(z_{\epsilon_0})$ whenever $|\epsilon - \epsilon_0| < \delta$. Then,

$$f_\epsilon(z) = f_{\epsilon_0}(z) + (\epsilon - \epsilon_0) \cdot g(z)$$

By Rouché's theorem, f_ϵ and f_{ϵ_0} have the same number of zeros inside $D_\zeta(z_{\epsilon_0})$. Hence, $z_\epsilon \in D_\zeta(z_{\epsilon_0})$ and so $|z_\epsilon - z_{\epsilon_0}| < \zeta$. $\square$

17. Let f be non-constant and holomorphic on an open set containing the closed disc.

(a) Show that if $|f(z)| = 1$ whenever $|z| = 1$ then the image of f contains the unit disc.

Proof. It suffices to show $f(z) = 0$ has a solution since for any other $w \in \mathbb{D}$, we can write $g(z) = f(z) + h(z)$ where $h(z) = -w$ and since on the unit circle $|f(z)| = 1 > |w| = |h(z)|$, by Rouché's theorem, we have that g and f have the same number of zeros. In particular, if f has a zero, then there exists z' such that $f(z') = w$ for every w in $\mathbb{D}$. On the other hand, if $f(z) = 0$ does not have a root, then $\frac{1}{f(z)}$ is holomorphic on $\mathbb{D}$. By Maximum Modulus principle,

$$\sup_{z \in \mathbb{D}} |f(z)| \leq 1 \implies |f(z)| \leq 1$$

and

$$\sup_{z \in \mathbb{D}} \frac{1}{|f(z)|} \leq 1 \implies 1 \leq |f(z)|$$

This shows $f(z)$ is constant on $\mathbb{D}$. Let Ω denote the domain of f , we can apply the maximum principle to $\Omega \setminus \mathbb{D}$ with f and $1/f$ to see that f is constant on $\Omega \setminus \mathbb{D}$ as well. Since f is holomorphic on Ω , we see that $f(z) = e^{i\theta}$ for some $\theta \in \mathbb{R}$. This contradicts with f being non-constant. Hence, $f(z) = 0$ has a root. And image of f contains the unit circle. $\square$

(b) If $|f(z)| \geq 1$ whenever $|z| = 1$ and if there exists $z_0 \in \mathbb{D}$ with $|f(z_0)| < 1$ then the image of f contains the disc.

Proof. Again, it suffices to show $f(z) = 0$ has a solution. If it doesn't then, $1/f(z)$ is holomorphic in $\mathbb{D}$ and so

$$\sup_{z \in \mathbb{D}} \frac{1}{|f(z)|} \leq 1 \implies 1 \leq |f(z)| \quad \forall z \in \mathbb{D}$$

But since $|f(z_0)| < 1$, this can't be true. $\square$

19. Prove the maximum principle for harmonic functions.

Proof. Let $u = \Re(f)$ attain a maximum at z_0 . Then the determinant $D = u_{xx}u_{yy} - u_{xy}^2 > 0$ is positive at $z_0 = (x_0, y_0)$. Using Cauchy Riemann equations, we can see

$$v_{xx}v_{yy} - v_{xy}^2 = -u_{yx} \cdot u_{xy} - (-u_{yy}) \cdot (u_{xx}) = u_{xx}u_{yy} - u_{xy}^2 > 0$$

which means v also attains a maximum at z_0 . Then, in a neighbourhood of z_0 , $|f(z)| = \sqrt{u^2(z) + v^2(z)} \leq |f(z_0)|$ so f is not open which is a contradiction. $\square$

20. (a) If f is holomorphic in the neighbourhood of the disc $D_r(z_0)$, show that for any $0 < s < r$ there exists a constant C depending on s and r such that

$$\|f\|_{L^\infty(D_s(z_0))} \leq C\|f\|_{L^2(D_r(z_0))}$$

Proof. By the mean value property,

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(z + r'e^{i\theta}) d\theta$$

for all $z \in D_r(z_0)$. Notice that

$$\begin{aligned} \int_{D_t(z)} f(x + iy) dx dy &= \int_0^t \int_0^{2\pi} f(z + ve^{i\theta}) v d\theta dv \quad \text{by Fubini.} \\ &= \int_0^t 2\pi f(z_0) v dv \\ &= \pi t^2 f(z_0) \end{aligned}$$

for all $z \in D_r(z_0)$ where t is such that $D_t(z) \subset D_r(z_0)$. Now by the Maximum Modulus Principle, for some $z' \in C_s(z_0)$, we have

$$\begin{aligned} \|f\|_{L^2(D_s(z_0))} &= |f(z')| \\ &= \frac{1}{\pi(r-s)^2} \left| \int_{D_{r-s}(z')} f(x + iy) dx dy \right| \\ &\leq \frac{1}{\pi(r-s)^2} \sqrt{\left(\int_{D_{r-s}(z')} |f(z)|^2 dx dy \right) \left(\int_{D_{r-s}(z')} 1 dx dy \right)} \quad \text{by Cauchy-Schwarz.} \\ &\leq \frac{1}{\sqrt{\pi}(r-s)} \sqrt{\int_{D_r(z_0)} |f(z)|^2 dx dy} \\ &= \frac{1}{\sqrt{\pi}(r-s)} \|f\|_{L^2(D_r(z_0))} \end{aligned}$$

as desired. $\square$

(b) If $\{f_n\}$ is a Cauchy sequence of holomorphic functions in the mean square norm $\|\cdot\|_{L^2(U)}$, then prove that $\{f_n\}$ converges uniformly to on every compact subset of U to a holomorphic function.

Proof. Let K be a compact subset of U . Let d be the distance between U^c and K . Since both are closed, $d > 0$. Then, by part a, we know that

$$\|f_n\|_{L^\infty(K)} \leq \frac{1}{\sqrt{\pi}d/2} \|f_n\|_{L^2(U)}$$

We want to show that for all $\varepsilon > 0$, there exists N such that for all $x \in K$,

$$|f_n(x) - f_m(x)| < \varepsilon$$

whenever $n, m > N$. Let $\varepsilon > 0$ be given. Then, there exists N such that $\|f_n - f_m\|_{L^2(U)} \leq \varepsilon$ whenever $n, m > N$. For such n, m and for all $x \in K$, we

have

$$\begin{aligned}
|f_n(x) - f_m(x)| &\leq \|f_n - f_m\|_{L^\infty(K)} \\
&\leq \frac{1}{\sqrt{\pi d}/2} \|f_n - f_m\|_{L^2(U)} \\
&\leq \frac{2}{\sqrt{\pi d}} \varepsilon
\end{aligned}$$

which shows f_n converges uniformly to a holomorphic function on compact subsets of U . $\square$

Chapter 8

1. Prove that a holomorphic map $f : U \rightarrow V$ is a local bijection on U if and only if $f'(z) \neq 0$ for all $z \in U$.

Proof. Let $f : U \rightarrow V$ be a local bijection. Let $z \in U$. Then, $f : D \rightarrow f(D)$ is a bijection for some neighbourhood of z . Then assume that $f'(z_0) = 0$ for some $z_0 \in D$. Let $w_0 = f(z_0)$ and consider $g(z) = f(z) - w_0$. Since f is holomorphic we can write

$$f(z) = a(z - z_0)^k + G(z)$$

where $k \geq 0$. Then, $g(z) = F(z) + G(z)$ where $F(z) = a(z - z_0)^k - w_0$. Since $|F(z)| > |G(z)|$ in a small neighbourhood of z_0 , we can apply Rouché's theorem to say that $g(z)$ has at least two zeros in a neighbourhood of z_0 . Because $F(z) = (z - z_0)^k a - w_0$ has k many zeros inside this neighbourhood. Then $g(z_1) = 0 = g(z_2) \implies f(z_1) = f(z_2) = w_0$, contradicting injectivity. So $f'(z) \neq 0$ for all $z \in U$. Conversely, if $f'(z) \neq 0$ for all $z \in U$, then f is locally invertible at every point by inverse function theorem. $\square$

2. **POTENTIAL** Suppose $F(z)$ is holomorphic near $z = z_0$ and $F(z_0) = F'(z_0) = 0$ while $F''(z_0) \neq 0$. Show that there are two curves Γ_1, Γ_2 that pass through z_0 , are orthogonal at z_0 , and so that F restricted to Γ_1 is real and has a minimum at z_0 while F restricted to Γ_2 is also real and has a maximum at z_0 .

Proof. Since $F''(z_0) \neq 0$ and $F'(z_0) = 0$, we can put $F(z) = g(z)^2$ near z_0 . Then,

$$F(z_0) = 0 \implies g(z_0) = 0$$

and

$$F''(z_0) = 2g'(z_0)^2 + 2g(z_0)g''(z_0) = 2g'(z_0)^2 \neq 0 \implies g'(z_0) \neq 0$$

which shows $g : D_r(z_0) \rightarrow D_\delta(0)$ is a diffeomorphism for some $r, \delta \in \mathbb{R}$. Since $F(z)$ is holomorphic we can write

$$g(z)^2 = (z - z_0)^2 H(z)$$

where $H(z)$ is a holomorphic function in a neighbourhood of z_0 . Then, by the maximum modulus principle,

$$\frac{|g(z)|^2}{|z - z_0|^2} \leq \frac{\delta^2}{r^2} \implies |g(z)| \leq |z - z_0| \frac{\delta}{r} \quad \forall z \in D_r(z_0)$$

Since $g(z)$ is a diffeomorphism, we can write

$$g^{-1}(z) = z_0 + a_1 z + a_2 z^2 + \dots$$

for all $z \in D_\delta(0)$. Then, $\frac{g^{-1}(z) - z_0}{z}$ is holomorphic on $D_\delta(0)$ and by the maximum modulus principle, we have

$$\frac{|g^{-1}(z) - z_0|}{|z|} \leq \frac{r}{\delta} \implies \frac{\delta}{r} |w - z_0| \leq |g(w)|$$

To use Schwarz's Lemma, we construct $G : \mathbb{D} \rightarrow \mathbb{D}$ by setting $G(z) = \frac{1}{\delta} g(rz + z_0)$. Then, $G(0) = 0$ and $|G(z)| = |z|$ so $G(z) = e^{i\theta} z$ for some $\theta \in \mathbb{R}$. In this case,

$$g(z) = \frac{\delta}{r} e^{i\theta} (z - z_0)$$

Let $\gamma_1(t) = z_0 + te^{-i\theta}$ and $\gamma_2(t) = z_0 + te^{i(\pi/2 - \theta)}$. This way we get

$$F(\gamma_1(t)) = \frac{\delta^2}{r^2} t^2 \quad F(\gamma_2(t)) = -\frac{\delta^2}{r^2} t^2$$

□

3. Suppose U and V are conformally equivalent. Prove that if U is simply connected, then so is V .

Proof. Let $\gamma_1, \gamma_2 : [0, 1] \rightarrow V$ be two curves with same end-points. Then define $\beta_i(t) = f^{-1}(\gamma_i(t))$ where $f : U \rightarrow V$ is a conformal map. Since U is simply connected, there exists a homotopy $F : [0, 1] \times [0, 1] \rightarrow U$ between β_1 and β_2 . Then, $f \circ F : [0, 1] \times [0, 1] \rightarrow V$ is a homotopy between γ_1 and γ_2 .

□

4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$?

Proof.

$$F(z) = \left[\left(i \frac{1-z}{1+z} \right) - i \right]^2$$

First map the disc into the upper half-plane then lower the upper half plane to $-i$ and upwards, then square it to get all possible complex numbers. □

5. Prove that $f(z) = -\frac{1}{2}(z + 1/z)$ is a conformal map from the upper half-disc into the upper half-plane.

Proof. It is obviously holomorphic. If $f(z) = f(z')$ then assuming $z \neq z'$,

$$z + \frac{1}{z} = z' + \frac{1}{z'} \implies zz' = 1$$

Then if $z = re^{i\theta}$ we must have $z' = \frac{1}{r}e^{i\theta}$. It is impossible for both z and z' to be on the upper half disc at the same time. Thus, f is injective. Let $w \in \mathbb{H}$, then

$$f(z) = w \implies z^2 + 2wz + 1 = 0$$

which has two roots if $w \neq 0$. If z_0 is a root of $f(z) - w = 0$ then by symmetry, $1/\bar{z}_0$ is the other root. This shows that at least one of the roots is inside the disc. Assume $z_0 \in \mathbb{D}$, then multiplying the last equation by $\bar{z}_0$ and equating the imaginary part to 0 gives

$$y = \frac{|z_0|^2 y'}{1 - |z_0|^2} > 0$$

where $z_0 = x + iy$ and $w = x' + iy'$, hence z_0 is in the upper half-disc. So f is surjective as well. $\square$

10. Let $F : \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function that satisfies $F(i) = 0$ and $|F(z)| < 1$. Prove that

$$|F(z)| \leq \left| \frac{z - i}{z + i} \right|$$

for all $z \in \mathbb{H}$.

Proof. Then, consider $g(z) = F(f(z))$ where $f : \mathbb{D} \rightarrow \mathbb{H}$ is given by $f(z) = i\frac{1-z}{1+z}$. We have $g(0) = 0$, by Schwarz' theorem,

$$|g(z)| \leq |z| \implies |F(i\frac{1-z}{1+z})| \leq |z| \implies |F(z)| \leq \left| \frac{i-z}{i+z} \right|$$

for all $z \in \mathbb{H}$ as desired.

11. We are given a holomorphic map $f : D_R(0) \rightarrow D_M(0)$. Let $\alpha = f(0)/M$ and define $G : \mathbb{D} \rightarrow \mathbb{D}$ be $G(z) = \varphi_\alpha \circ \frac{1}{M}f(Rz)$. Notice $G(0) = 0$, then apply Schwarz lemma to get

$$\frac{|f(0) - f(z)|}{|M^2 - \bar{f}(0)f(z)|} \leq \frac{|z|}{R}$$

$\square$

12. (a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is analytic and has two fixed points then f is the identity.

Proof. Let $f(z_1) = z_1, f(z_2) = z_2$ be the two fixed points. Consider $g = \varphi_{z_1} \circ f \circ \varphi_{z_1} : \mathbb{D} \rightarrow \mathbb{D}$. We have $g(0) = 0$. Let α be such that $\varphi_{z_1}(\alpha) = z_2$. Then,

$$g(z_2) = z_2$$

By Schwarz lemma, $g = Id$ then $f = Id$. $\square$

(b) Must every holomorphic function have a fixed point?

Proof. No, not every function $f : \mathbb{D} \rightarrow \mathbb{D}$ has a fixed point. Consider $f(z) = F^{-1} \circ H \circ F(z)$ where $F(z) = i \frac{1-z}{1+z}$ and $H(z) = z - 1$.

$$f(z) = z \implies H \circ F(z) = F(z)$$

which is impossible. □

13. The pseudo-hyperbolic distance is given by

$$\rho(z, w) = \frac{|w - z|}{|1 - \bar{w}z|}$$

(a) Prove that if $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic then

$$\rho(f(z), f(w)) \leq \rho(z, w)$$

for all $z, w \in \mathbb{D}$. Moreover, prove that if f is an automorphism, then it preserves the pseudo-hyperbolic distance.

Proof. Consider the function $g = \varphi_{f(w)} \circ f \circ \varphi_w : \mathbb{D} \rightarrow \mathbb{D}$ which fixes the origin and is holomorphic. Applying Schwarz lemma gives

$$\rho(f(w), f(z)) \leq \rho(z, w)$$

as desired. If f is an automorphism, the same inequality is true for f^{-1} hence we get

$$\rho(f(w), f(z)) = \rho(z, w)$$

□

(b) Prove that

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}$$

for all $z \in \mathbb{D}$.

Proof. For the second part, if we put $w = z + h$ and let $h \rightarrow \infty$, then we get

$$\frac{|f'(z)|}{|1 - |f(z)|^2|} \leq \frac{1}{1 - |z|^2}$$

□

14. **POTENTIAL** Prove that every conformal mapping of the upper half plane to the unit disc take the form

$$f(z) = e^{i\theta} \frac{z - \beta}{z - \bar{\beta}}, \quad \theta \in \mathbb{R}, \beta \in \mathbb{H}$$

Proof. Let $f : \mathbb{H} \rightarrow \mathbb{D}$ be a conformal map. Then there exists $\beta \in \mathbb{H}$ with $f(\beta) = 0$. Define $g(z) = \frac{z\bar{\beta}-\beta}{z-1}$ for all $z \in \mathbb{D}$. Notice that g has an inverse $g^{-1}(z) = \frac{z-\beta}{z-\bar{\beta}}$. Since $g \circ g^{-1}(z) = z$ for all $z \in \mathbb{H}$, we see that $g : \mathbb{D} \rightarrow \mathbb{H}$ is a conformal map. Then

$$f \circ g : \mathbb{D} \rightarrow \mathbb{D}$$

is an automorphism of $\mathbb{D}$ with $f \circ g(0) = f(\beta) = 0$. Then, by Schwarz lemma $|f \circ g(z)| \leq |z|$. Since this map is an automorphism, the same inequality is true for its inverse and so $|f(g(z))| = |z|$ so $f(g(z)) = e^{i\theta}z$ for some $\theta \in \mathbb{R}$. Replacing $g(z)$ with z , we obtain

$$f(z) = e^{i\theta} \frac{z - \beta}{z - \bar{\beta}}$$

□

15. (a) Suppose Φ is an automorphism of $\mathbb{H}$ that fixes three distinct points on the real axis. Then Φ is the identity.

Proof. If Φ is a automorphism of the upper half-plane then $\Phi = f_M$ for some $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$

$$\frac{az+b}{cz+d} = z \implies cz^2 + (d-a)z - b = 0$$

This equation can have at most 2 distinct solutions, if Φ has three fixed points then we must have $c = 0, a = d, b = 0$. □

(b) Suppose x_1, x_2, x_3 and y_1, y_2, y_3 are two pairs of distinct points on the real axis with

$$x_1 < x_2 < x_3 \quad y_1 < y_2 < y_3$$

Prove that there exists a unique automorphism Φ of $\mathbb{H}$ so that $\Phi(x_j) = y_j$.

Proof. Say we want to find a map that sends x_1, x_2, x_3 to y_1, y_2, y_3 . First find a map f_M that sends x_1, x_2, x_3 to $0, 1, \infty$ and another map f_N that sends y_1, y_2, y_3 to $0, 1, \infty$ then $f_N^{-1} \circ f_M = f_{N^{-1}M}$ sends x_1, x_2, x_3 to y_1, y_2, y_3 . If you write out the equations with the determinant restriction, then you will see that

$$M = \frac{1}{(x_1 - x_2)(x_2 - x_3)(x_3 - x_1)} \begin{pmatrix} x_2 - x_3 & x_1(x_3 - x_2) \\ x_2 - x_1 & x_3(x_1 - x_2) \end{pmatrix}$$

And N is a similar matrix. □

16) Let $f(z) = \frac{i-z}{i+z}$ and $f^{-1}(z) = i \frac{1-z}{1+z}$. Given $\theta \in \mathbb{R}$ find real numbers a, b, c, d such that

$$\frac{az+b}{cz+d} = f^{-1}(e^{i\theta}f(z))$$

Proof.

$$\begin{aligned} f^{-1}(e^{i\theta} f(z)) &= \frac{(e^{i\theta} - 1) + (1 + e^{i\theta})iz}{(1 + e^{i\theta})i + (1 - e^{i\theta})z} \\ &= \frac{1 + tz}{t - z} \end{aligned}$$

where

$$t = i \frac{e^{i\theta} + 1}{e^{i\theta} - 1} = \frac{2 \sin \theta}{|e^{i\theta} - 1|^2} \in \mathbb{R}$$

Then the matrix is

$$M = \frac{1}{t^2 + 1} \begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix}$$

□